

Use Case Flow — Smart Complaint Management System

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Each use case below lists its ID, name, actor, precondition, main flow, one alternative flow, and postcondition. Alternative flows are labelled with the main-flow step number they branch from, followed by "a" (e.g. 7a branches from step 7).

UC-01 - Submit Complaint

Use Case ID	UC-01
Use Case Name	Submit Complaint
Actor	Citizen
Precondition	Citizen must be logged in.
Main Flow	1. Citizen taps "New Complaint". 2. Fills in title, description, and severity. 3. System suggests top-3 categories via NLP. 4. Citizen accepts or overrides the suggested category. 5. Attaches images and enables GPS (optional). 6. Citizen taps "Submit". 7. System validates all mandatory fields. 8. System generates a unique ticket ID. 9. System routes the complaint to the appropriate department/agent. 10. System sends a confirmation notification with the ticket ID.
Alternative Flow	7a. If mandatory fields are missing, system displays validation errors; citizen corrects the fields and resubmits.
Postcondition	Complaint ticket is created, stored, and routed to the appropriate department.

UC-02 - Track Complaint Status

Use Case ID	UC-02
Use Case Name	Track Complaint Status
Actor	Citizen
Precondition	Citizen must be logged in and have at least one submitted complaint.
Main Flow	1. Citizen navigates to "My Complaints". 2. Selects a complaint from the list. 3. System displays current status and timeline. 4. Citizen views status history with timestamps.
Alternative Flow	2a. If the complaint ID is invalid or deleted, system displays an error.
Postcondition	Citizen views the up-to-date status of the selected complaint.

UC-03 - Rate Resolution

Use Case ID	UC-03
Use Case Name	Rate Resolution
Actor	Citizen
Precondition	Complaint status must be "Resolved".
Main Flow	1. Citizen opens the resolved complaint. 2. Selects a star rating (1-5). 3. Enters an optional comment. 4. Submits the feedback. 5. System saves the rating and updates the CSAT score.
Alternative Flow	4a. If rating submission fails, system prompts the citizen to retry.
Postcondition	Rating and feedback are recorded; complaint moves to "Closed".

UC-04 - Resolve Complaint

Use Case ID	UC-04
Use Case Name	Resolve Complaint
Actor	Agent
Precondition	Complaint must be assigned to the agent.
Main Flow	1. Agent opens the assigned complaint. 2. Reviews complaint details, attachments, and notes. 3. Performs the required action (field visit, coordination, fix). 4. Adds resolution notes describing the action taken. 5. Marks the complaint as "Resolved". 6. System notifies the citizen.
Alternative Flow	3a. If more information is needed, agent sets status to "Pending Info" and queries the citizen.
Postcondition	Complaint is marked "Resolved" and the citizen is notified.

UC-05 - Add Notes and Evidence

Use Case ID	UC-05
Use Case Name	Add Notes and Evidence
Actor	Agent
Precondition	Agent must be assigned to the complaint.
Main Flow	1. Agent opens the complaint. 2. Selects "Add Note". 3. Enters note text describing progress or findings. 4. Attaches evidence file (optional, PDF/image, max 10 MB). 5. Saves the note.
Alternative Flow	4a. If the attachment exceeds 10 MB, system rejects the file.
Postcondition	Note and evidence are stored in the complaint's resolution trail.

UC-06 - Escalate Complaint

Use Case ID	UC-06
Use Case Name	Escalate Complaint
Actor	Supervisor
Precondition	Complaint must be stalled beyond SLA or flagged for review.
Main Flow	1. Supervisor opens the "Overdue" tab. 2. Selects the stalled complaint. 3. Clicks "Escalate". 4. Enters an escalation reason and selects the senior team. 5. System moves the ticket to the priority queue and resets the SLA timer. 6. System notifies the original agent, new team, and citizen.
Alternative Flow	4a. If no senior team is configured for the category, system alerts the admin and allows manual assignment.
Postcondition	Complaint is reassigned to the priority queue with a new SLA; all parties are notified.

UC-07 - Auto-Categorise Complaint

Use Case ID	UC-07
Use Case Name	Auto-Categorise Complaint
Actor	System (invoked by Citizen/Agent)
Precondition	Complaint description text must be available.
Main Flow	1. System receives the complaint description text. 2. NLP model processes the text and ranks the top-3 categories. 3. System returns confidence scores for each suggestion. 4. Citizen or agent accepts or overrides the suggestion.
Alternative Flow	3a. If confidence is below the 70% threshold, system requests manual category selection.
Postcondition	Complaint is tagged with a category before routing.

UC-08 - View Dashboard

Use Case ID	UC-08
Use Case Name	View Dashboard
Actor	Supervisor, Admin
Precondition	User must have Supervisor or Admin role.
Main Flow	1. User navigates to "Dashboard". 2. System loads KPIs (open, resolved, overdue, avg. resolution time). 3. User applies filters (date range, department, agent). 4. System refreshes the displayed data every 60 seconds.
Alternative Flow	2a. If no data matches the selected filters, system displays an empty-state message.
Postcondition	User views up-to-date performance metrics for the selected filters.

UC-09 - Manage Users

Use Case ID	UC-09
Use Case Name	Manage Users
Actor	Admin
Precondition	Admin must be logged in.
Main Flow	1. Admin navigates to "User Management". 2. Clicks "Add User". 3. Enters name, email, department, and role. 4. Saves the account. 5. System sends a welcome email to the new user.
Alternative Flow	3a. If the email already exists, system displays a duplicate-account error.
Postcondition	New user account is created with the correct role permissions.

UC-10 - Configure SLA Rules

Use Case ID	UC-10
Use Case Name	Configure SLA Rules
Actor	Admin
Precondition	Admin must be logged in.
Main Flow	1. Admin navigates to "SLA Configuration". 2. Selects a complaint category and severity level. 3. Sets response-time and resolution-time targets. 4. Saves the rule.
Alternative Flow	3a. If entered values are invalid (e.g. negative time), system rejects the configuration.
Postcondition	SLA rule is updated and applied to new complaints immediately.